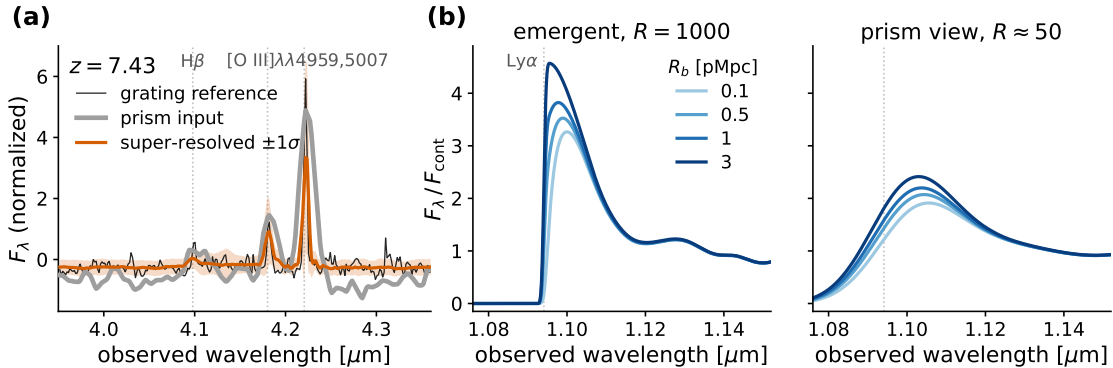


Deep Learning for Galaxy Spectral Super-Resolution

Aryana Haghjoo

Modern infrared telescopes like JWST and Nancy Grace Roman enable cosmic observations [Gardner et al., 2023], but efficient spectroscopic modes operate at low resolution: JWST’s NIRSpec prism ($R \sim 100$) [Jakobsen et al., 2022] and Roman’s grism. Spectral features encoding critical physics blur below instrumental limits. I develop physics-informed deep learning frameworks that super-resolve low-resolution galaxy spectra by $\sim 10\times$, recovering hidden astrophysical information [Haghjoo et al., 2026].

Using paired JWST/NIRSpec observations from JADES [Eisenstein et al., 2026] (same galaxies in high-resolution and prism modes), I train generative models with physics-informed loss functions [Ledig et al., 2017, Vaswani et al., 2017, He et al., 2016]. Figure 1(a) demonstrates super-resolution on a $z = 7.43$ galaxy, recovering features invisible in prism data. Figure 1(b) shows the astrophysical impact: Lyman-alpha profiles encode ionized bubble sizes (R_b) around galaxies. High-resolution profiles are distinguishable; prism-resolution profiles appear identical. Our method recovers this hidden information, enabling physical measurements from scientifically limited data.



Summer 2027 at IPAC: The fellowship focuses on four integrated components: (1) validating recovery of astrophysical diagnostics using held-out JADES data, comparing deep learning against classical deconvolution methods [Lucy, 1974, Richardson, 1972, Scourfield et al., 2023]; (2) adapting the framework to Roman grism with IPAC’s mission team; (3) developing multi-modal models; (4) extending to Euclid spectroscopy. Timing is critical: Roman’s first data release is expected Summer 2027. On-site access to real mission data and Dr. Shoubaneh Hemmati—active member of Euclid and Roman missions—enables immediate validation impossible remotely. In-person collaboration ensures our methodology aligns with mission requirements and delivers forecasts to inform survey strategy.

This work advances galaxy astrophysics while demonstrating how machine learning extracts maximum value from observational data [Schawinski et al., 2017, Fabbro et al., 2018]. Open-

source code and models will enable researchers to apply spectral super-resolution across datasets, extending the science reach of JWST, Roman, Euclid, and future missions.

References

- Daniel J Eisenstein, Chris Willott, Stacey Alberts, et al. Overview of the jwst advanced deep extragalactic survey (jades). *The Astrophysical Journal Supplement Series*, 283(1):6, 2026.
- S. Fabbro, K. A. Venn, et al. An application of deep learning in the analysis of stellar spectra. *Monthly Notices of the Royal Astronomical Society*, 475:2978–2993, 2018.
- Jonathan P. Gardner, John C. Mather, et al. The James Webb Space Telescope mission. *Publications of the Astronomical Society of the Pacific*, 135:068001, 2023.
- Aryana Haghighi, Shoubaneh Hemmati, Bahram Mobasher, Nima Chartab, Alexander de la Vega, Tim Eifler, Emily Ramey, Hooshang Nayyeri, and Zahra Sattari. Learning to see sharper: A physics-informed artificial intelligence framework for super-resolving galaxy spectra. *The Astrophysical Journal*, 2026. submitted.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.
- P. Jakobsen, P. Ferruit, et al. The near-infrared spectrograph (nirspec) on the James Webb Space Telescope. *Astronomy & Astrophysics*, 661:A80, 2022.
- Christian Ledig, Lucas Theis, Ferenc Huszár, et al. Photo-realistic single image super-resolution using a generative adversarial network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4681–4690, 2017.
- L. B. Lucy. An iterative technique for the rectification of observed distributions. *The Astronomical Journal*, 79:745, 1974.
- William Hadley Richardson. Bayesian-based iterative method of image restoration. *Journal of the Optical Society of America*, 62:55–59, 1972.
- Kevin Schawinski, Ce Zhang, Hantian Zhang, et al. Generative adversarial networks recover features in astrophysical images of galaxies. *Monthly Notices of the Royal Astronomical Society: Letters*, 467:L110–L114, 2017.
- M. Scourfield, A. Saintonge, et al. De-noising of galaxy optical spectra with autoencoders. *Monthly Notices of the Royal Astronomical Society*, 526:3037–3050, 2023.
- Ashish Vaswani, Noam Shazeer, et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.